# TIMOTHY LOCK

me@timothylock.ca | timlock.dev | Seattle, WA

## PROFESSIONAL SUMMARY

Backend and platform engineer with 7+ years of experience building distributed systems, Kubernetes platforms, cloud infrastructure, and high-throughput microservices. Proven record of operating systems at scale across AWS, GCP, Azure, and on-premises environments, including platforms serving 90 million users and more than 80,000 requests per second. Combines hands-on Go development with platform engineering, GitOps, API gateway architecture, observability, and technical leadership to improve reliability, security, and developer velocity.

## TECHNICAL SKILLS

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| **Languages:** | Go (Golang), PHP, Java, Python, SQL, JavaScript, HTML |
| **Cloud & Infrastructure:** | AWS, GCP, Azure, Kubernetes, Docker, Terraform, Helm, Argo CD, GitOps |
| **Backend & APIs:** | Microservices, distributed systems, REST APIs, Kong API Gateway, OAuth 2.0 |
| **Databases & Storage:** | PostgreSQL, MySQL, DynamoDB, Redis, Amazon Aurora, RDS, SQS/SNS |
| **Tools & Observability:** | Git, GitHub, Jira, PagerDuty, CI/CD, CircleCI, Grafana, Datadog, Dynatrace, Linux, Windows |

## PROFESSIONAL EXPERIENCE

**Electronic Arts** — Seattle, WA

*Software Engineer* — October 2023 - Present

- Optimized game-development pipelines, cutting annual cost from ~$144,000 to under $120 and data from ~2 PB to 2 TB.
- Build Kubernetes operators in Go with Kubebuilder to automate cluster lifecycle and resource orchestration.
- Deliver GitOps with Argo CD and Helm across hundreds of clusters spanning AWS, GCP, Azure, and on-prem.
- Design shared infrastructure APIs and CLIs that give game studios consistent access to platform services.
- Build AI infrastructure for developer tooling, using vector-based systems to improve pipeline efficiency and reliability.

**Splashthat** — Toronto, ON

*Software Engineer* — September 2021 - October 2023

- Optimized Kong API Gateway and OAuth 2.0 infrastructure for platform-wide security and performance.
- Built rate limiting that blocked 5,000+ malicious requests per second while preserving legitimate traffic.
- Created a reusable Go microservice template that accelerated migration to a standardized service architecture.
- Built event-driven API integrations for Salesforce AppExchange, Marketo, HubSpot, and other enterprise platforms.
- Improved high-traffic API response times by 62% through SQL query analysis and optimization.
- Led agile ceremonies and roadmap planning while mentoring junior engineers.

**Wattpad** — Toronto, ON

*Software Engineer* — June 2019 - September 2021

- Built backend microservices handling 80,000+ requests per second for a platform serving 90 million users.
- Migrated legacy PHP monoliths to Go microservices backed by Redis, MySQL, and Amazon Aurora.
- Built image moderation processing 3M+ images monthly and removing prohibited content.
- Led Sign in with Apple and maintained OAuth integrations with Google, Facebook, and other identity providers.
- Developed services for Wattpad Premium and Paid Stories, supporting subscriptions and monetization.

**Wattpad** — Toronto, ON

*Associate Software Engineer* — May 2017 - September 2018

- Developed ad-exchange integrations and backend features for a platform serving millions of users each day.
- Collaborated with senior engineers on system design, implementation, testing, and code reviews.

**University of Toronto - Information & Technology Services** Toronto, ON

*IT Helpdesk Analyst* September 2018 - May 2019

- Provided technical support to more than 1,150 students across the university network.
- Designed solutions that integrated new users into 15-year-old legacy infrastructure.
- Upgraded campus networking equipment to improve overall network performance and reliability.

## EDUCATION

**University of Toronto** Toronto, ON

*Honours Bachelor of Science - Computer Science Specialist* September 2014 - April 2019

## SELECTED PROJECTS

**go-signin-with-apple** Open Source Go Library

- Open-source Go library for Apple's Sign in with Apple REST API: token validation and revocation, ID token claims parsing, user migration across developer teams, and server-to-server event notifications. 245+ GitHub stars and 75 forks.

## CERTIFICATIONS

- Certified Kubernetes Application Developer (CKAD), Linux Foundation, 2024.